Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (West), Mumbai-400058-India
(Autonomous Institute Affiliated to University of Mumbai)

Department of Computer Engineering

Generative AI Lab Manual AY 2026-27

Class: BTech(CE) | **Semester: VII**
Course Name: Generative AI (PE III) | **Course Code: CE413**
Faculty Name: Prof. Yogita Patil | **Academic Year: 2026-27**

Experiment 10: Multimodal Content Generation Using Generative AI

1. Aim

To generate an image and music using a common text prompt, and produce a short video by integrating the generated image, music, and text or motion effects using lightweight generative AI models and video-processing tools.

2. Learning Outcomes

After completing this experiment, students will be able to:

- Understand text-to-image and text-to-music generation.
- Design prompts that maintain a consistent theme across multiple modalities.
- Integrate generated visual and audio content into a short video.
- Evaluate the quality, consistency, and limitations of AI-generated content.
- Identify ethical concerns, representation bias, copyright, and responsible AI practices.

3. Software and Resources

Google Colab with GPU runtime where available.

Hugging Face model repository.

Python-based image and audio processing libraries.

A video editor or video-processing library for final composition.

Google Drive for saving generated outputs.

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4. Problem Statement

Create a 15–30-second educational awareness video on **"Human Genetic Diversity and Responsible AI in Healthcare."** Generate an image and a music track from a shared theme, then combine them into a video with suitable titles, transitions, and motion effects.

The video must communicate that AI can support scientific research but cannot establish a genetic diagnosis from facial appearance alone.

6. Models and Tools

Task	Recommended model/tool	Purpose
Text-to-image	stabilityai/sd-turbo	Generate an image quickly using a text prompt
Text-to-music	facebook/musicgen-small	Generate a short instrumental track from a text description
Video composition	MoviePy or FFmpeg	Combine the image, music, titles, and motion effects
Optional image animation	A pretrained image-to-video model, if sufficient GPU resources are available	Create motion from the generated image

Model repositories:

SD-Turbo: https://huggingface.co/stabilityai/sd-turbo

MusicGen Small: https://huggingface.co/facebook/musicgen-small

Important: SD-Turbo is designed for fast image synthesis. MusicGen Small is a 300-million-parameter music model, but it can still require considerable GPU memory. The final video does not require a separate video-generation model; students can create it by animating the generated image and synchronizing it with the generated music.

6. Procedure

Task 1: Set up Google Colab

1) Open Google Colab and create a new notebook.
2) Select a GPU runtime if one is available.
3) Check the available GPU memory and storage.

4) Install and import the required libraries using the supplied instructor notebook.

5) Create a project folder to store all generated files.

Task 2: Design the common prompt

I. Prepare a prompt describing the theme, visual style, atmosphere, and intended message.

II. Reuse the same core theme for image and music generation.

III. Record the complete prompts used for each modality.

IV. Modify the descriptive details when necessary without changing the central theme.

Example common prompt:

"Human genetic diversity and responsible AI in healthcare, represented through respectful illustrations of diverse human faces, abstract DNA structures, and subtle AI-inspired scientific elements. Use a futuristic blue-and-purple visual style with a calm, empathetic, hopeful atmosphere."

Task 3: Generate the image

Load the pretrained SD-Turbo model.

Enter the common prompt.

Generate an image at a moderate resolution suitable for a short video.

Inspect the image for visual quality, relevance, unwanted artefacts, and respectful representation.

If time permits, generate a second version by modifying the prompt.

Save the selected image in PNG or JPG format.

Record the model name, prompt, image dimensions, generation time, and observations.

Task 4: Generate the music

Load the pretrained MusicGen Small model.

Create a music prompt based on the same theme.

Specify the genre, mood, instruments, and desired duration.

Generate a short instrumental track.

Listen to the audio and check whether it matches the intended atmosphere.

If time permits, compare two music prompts and select the more suitable result.

Save the selected track in WAV or another supported audio format.

Example music prompt:

"Gentle, inspirational cinematic instrumental music about scientific discovery and human diversity, with soft piano, ambient synthesizers, a gradual uplifting progression, and a calm, hopeful mood."

Record the prompt, audio duration, generation time, and observations.

Task 5: Create the video

Load the saved image and audio file into the video-composition workflow.

Set the target video duration to 15–30 seconds.

Apply a slow zoom, pan, or other subtle motion effect to the image.

Add an opening title: "Human Genetic Diversity and Responsible AI."

Add one or two short messages, such as:"Every individual is unique."

"AI can support research, but clinical diagnosis requires qualified professionals."

Add suitable transitions and ensure that the text remains readable.

Synchronize the music with the video duration and adjust the audio level.

Export the final video in MP4 format.

Play the exported video and verify that the image, audio, titles, and transitions work correctly.

Students may use MoviePy or FFmpeg in Colab for this task. A separate image-to-video model is optional, not mandatory.

Task 6: Evaluate the results

Evaluate the image, music, and final video separately and as a combined output.

a) Does the image match the prompt?

b) Does the music convey the intended mood?

c) Does the video maintain consistency between its visual and audio components?

d) Are the text, transitions, and timing appropriate?

e) Did the models introduce unwanted visual or audio artefacts?

f) Are the generated representations respectful and free from unsupported medical claims?

Task 7: Save the deliverables

7. Expected Output

A short educational awareness video integrating an AI-generated image, AI-generated instrumental music, and appropriate text or motion effects.

8. Ethical Considerations

Do not upload identifiable patient photographs or sensitive medical information.

Do not label generated facial appearances as evidence of a genetic disorder.

Do not present synthetic images as genuine patient cases or clinical evidence.

Consider demographic representation bias in generated images.

Check model licences and usage restrictions before publishing or distributing outputs.

Clearly disclose that the content is AI-generated.

9. Conclusion

Students will demonstrate a multimodal generative AI workflow by creating an image, generating matching music, integrating both into a short video, and critically evaluating the resulting content.